HEXACORE
PRECISION TECHNOLOGIES

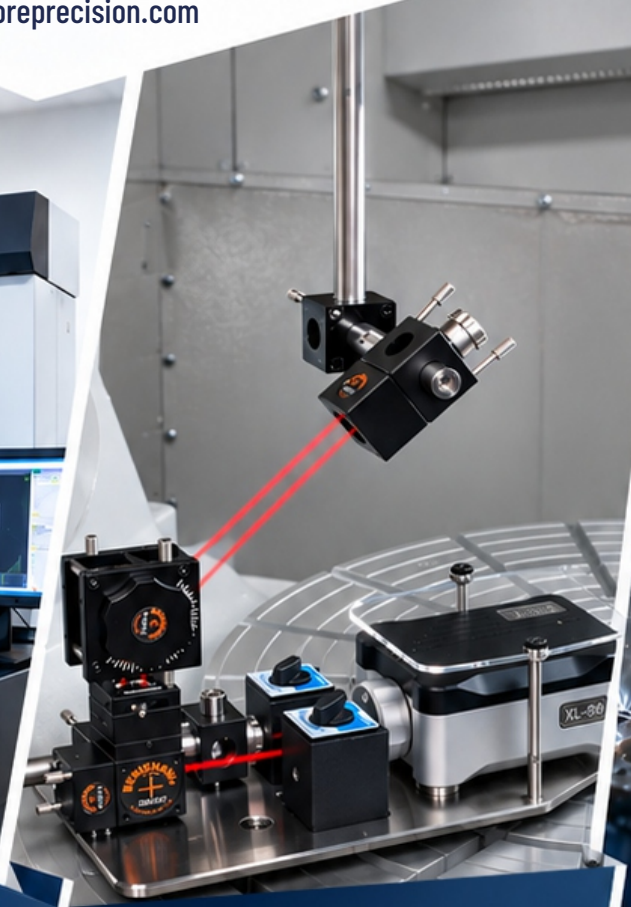
www.hexacoreprecision.com

MEASURE | MANUFACTURE | EXCEL

PRECISION TODAY
A BETTER TOMORROW

PRECISION
PEOPLE
PROCESS
PERFORMANCE







- CNC Machine Calibration & analyses
- CMM analyses
- RetroFit & Re-Installation
- Probe Installation

- Accuracy Verification
- Laser Alignment Check
- Measurement Error Correction
- Calibration Report & Certification

- CNC Machine Geometric Accuracy Inspection
- Straightness Measurement
- Spindle Alignment & Runout Inspection
- Machine Leveling & Relocation
- Machine Retrofit

LASER CALIBRATION | CNC SERVICES | AUTOMATION | METROLOGY | PROBING SOLUTIONS

PRECISION
FOR A SMARTER
WORLD

QUALITY EXPERTISE INNOVATION PARTNERSHIP

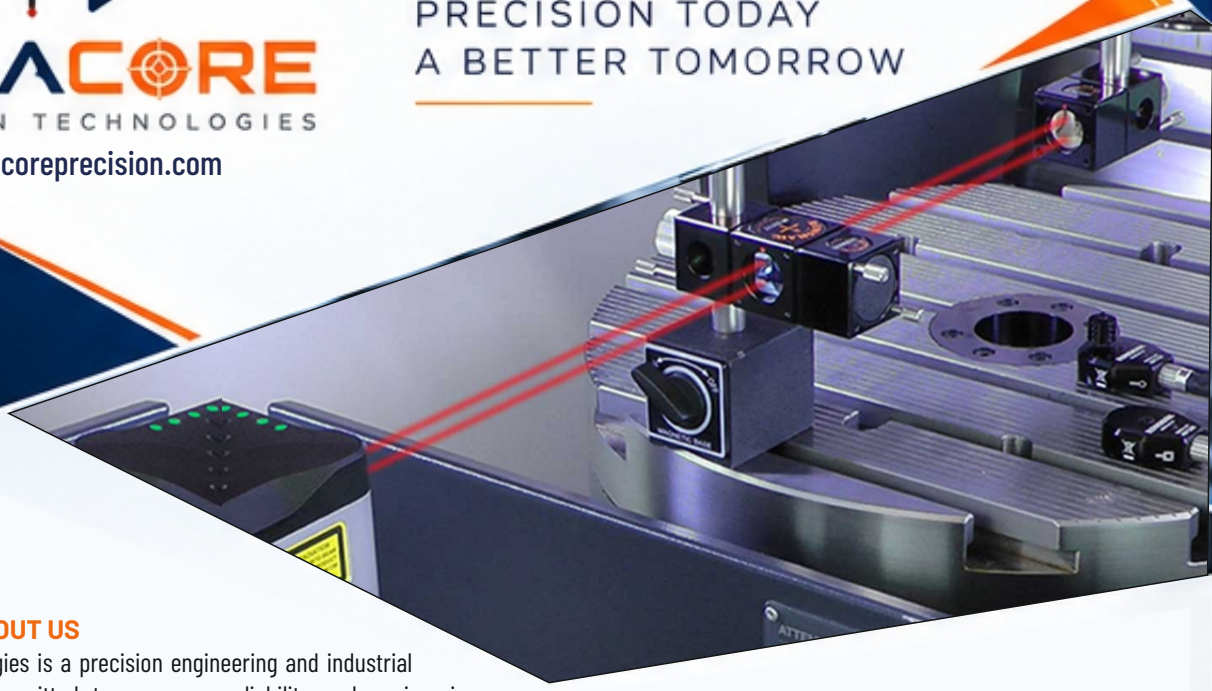
HEXACORE
PRECISION TECHNOLOGIES

www.hexacoreprecision.com

MEASURE | MANUFACTURE | EXCEL

PRECISION TODAY
A BETTER TOMORROW

ENGINEERING
CONFIDENCE
THROUGH
PRECISION



ABOUT US

Hexacore Precision Technologies is a precision engineering and industrial metrology solutions company committed to accuracy, reliability and engineering excellence.

We provide specialized solutions for CNC machines, CMMs, machine tools and measuring instruments, helping industries improve dimensional accuracy, equipment performance, productivity and manufacturing quality.

Our approach focuses on identifying the root cause of errors, implementing the right corrective action and verifying equipment performance with measurable results.

OUR FOUNDATION

PRECISION

Accurate measurement and controlled machine performance are the foundation of quality manufacturing.

ENGINEERING

We identify the source of errors and provide practical engineering solutions.

TRACEABILITY

We follow documented procedures and reliable measurement practices for consistent results.

RELIABILITY

Our goal is dependable, long-term machine and measurement performance.

PARTNERSHIP

We build lasting technical relationships through diagnosis, correction, verification and continued support.

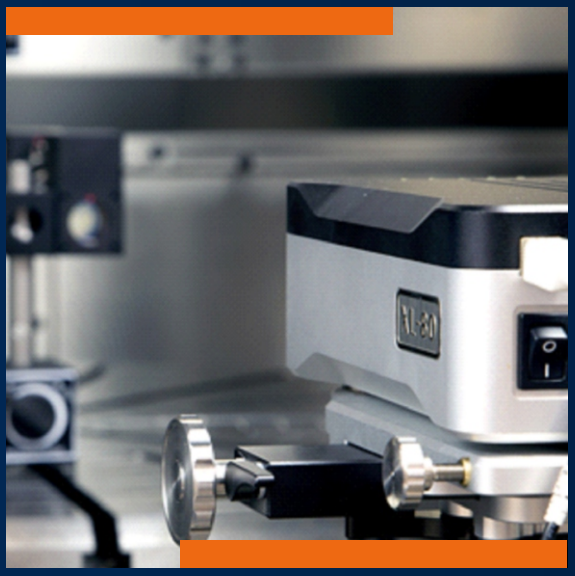
OUR APPROACH

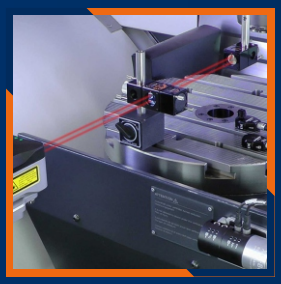


PRECISION
FOR A SMARTER
WORLD

analysesS

1. LINEAR LASER CALIBRATION (upto 40 Meters)
2. MACHINE HEALTH DIAGNOSE (BALLBAR TEST)
3. ROTARY AXIS CALIBRATION
4. OFF AXIS ROTARY CALIBRATION
5. AXIS STRAIGHTNESS TESTING
6. PREVENTIVE MAINTENANCE CONTRACT





LINEAR LASER CALIBRATION : Ensures precise positioning accuracy of machine axes using laser interferometer systems. Specs/Attributes: Accuracy up to 0.005 ppm, positioning & repeatability analysis, pitch error compensation, ISO 230-2 compliant.



Machine Health Diagnose (Ballbar Test) : Evaluates overall CNC performance by analyzing circular interpolation errors. Specs/Attributes: Detects backlash, servo mismatch, cyclic errors; 360° and 220° tests; quick diagnostic within minutes; ISO 230-4 standard.



Rotary Axis Calibration : Calibrates rotary axes for accurate angular positioning and indexing. Specs/Attributes: Angular accuracy up to +1 arc-sec, indexing error mapping, encoder verification, improves 4th/5th axis performance.



Off-Axis Rotary Calibration : Measures rotary axis errors when positioned away from the center line. Specs/Attributes: Eccentricity and tilt error analysis, volumetric compensation support, critical for multi-axis machining accuracy.



Preventive Maintenance Contract : Scheduled maintenance to ensure consistent machine performance and reliability. Specs/Attributes: Periodic calibration checks, lubrication inspection, alignment verification, customized analyses intervals, downtime reduction.

TYPES OF MACHINE WE CALIBRATE



CNC Turning Centers (Lathes)



Vertical Machining Centers (VMC)



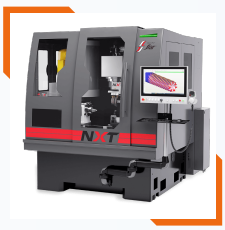
Horizontal Machining Centers (HMC)



5-Axis CNC Machines



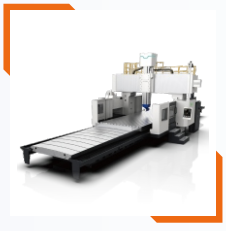
EDM Machines
(Wire Cut & Die Sinker)



Tool & Cutter
Grinding Machines



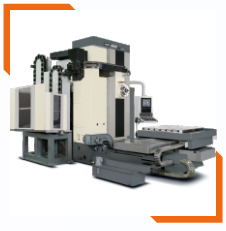
Gantry / Plano Milling Machines



Double Column
Machining Centers



CNC Grinding Machines
(Surface & Cylindrical)



Boring Machines
(Horizontal & Vertical)



Gear Hobbing /
Gear Cutting Machines



Special Purpose
Machines (SPM)

CNC CONTROLERS WE SUPPORT

FANUC | SIEMENS (SINUMERIK) | HEIDENHAIN | MITSUBISHI | MAZATROL (MAZAK) | HAAS | FAGOR | SYNTEC | DELTA | OKUMA (OSP)

HURCO (WINMAX) | YASKAWA | EXCETEK | NTC

HEXACORE
PRECISION TECHNOLOGIES
www.hexacoreprecision.com

ACCURACY
DRIVES
PROGRESS

Probing Solutions

Complete CNC probing solutions for accurate workpiece setting, tool measurement, inspection and process control, helping improve machining accuracy, reduce setup time and minimize operator errors.

Probe Installation

Professional installation, configuration and calibration of CNC touch probes and tool-setting systems to ensure reliable measurement and accurate machine integration.

analyses & Troubleshooting

Inspection, diagnosis and repair of probing-system issues including signal errors, calibration problems, communication faults and probe malfunction, restoring dependable measurement performance.













CNC Machine Geometric Accuracy Inspection

Comprehensive inspection of CNC machine geometry to verify positional accuracy, alignment, straightness, level and overall machine performance.

Straightness Measurement

Checks the linear straightness of machine axes to identify deviation, wear and alignment errors that may affect machining accuracy.

Spindle Alignment & Runout Inspection

Measures spindle alignment and rotational runout to ensure smooth rotation, dimensional accuracy, better surface finish and longer tool life.

Machine Leveling & Relocation

Precision leveling and alignment of machines during installation or relocation to maintain stability, geometry and consistent machining performance.

Machine Retrofit

Upgrading and modernizing existing machines with improved control systems, electrical components, drives, measuring systems and mechanical enhancements to improve accuracy, reliability, productivity and machine life.



HEXACORE
PRECISION TECHNOLOGIES
www.hexacoreprecision.com

ACCURACY DRIVES PROGRESS



1D & 2D HEIGHT GAUGES

1D & 2D Height Gauges are precision measuring instruments designed for accurate dimensional inspection of components on a surface plate. They are widely used in manufacturing, tool rooms, quality control, and metrology applications.

1D Height Gauge: Used for precise vertical measurements such as height, step height, depth, diameter, and center-to-center dimensions.

2D Height Gauge: Provides advanced measurement in both X and Z directions, enabling measurement of heights, diameters, distances, center locations, angles, and other geometric features.

Offers high accuracy, repeatability, and easy operation for inspection of machined components.

Suitable for production inspection, calibration laboratories, automotive, aerospace, and precision engineering industries.



VIDEO MEASURING MACHINE (VMM)

A Video Measuring Machine (VMM) is a high-precision, non-contact optical inspection system used to measure the dimensions and geometric features of components using a high-resolution camera, optics, lighting, and measurement software.

It is ideal for measuring small, delicate, complex, or precision components where conventional contact measuring methods may be difficult or unsuitable.

Measures length, width, diameter, radius, angle, pitch, distance, and profile

Provides fast and accurate non-contact measurement

High-resolution camera with advanced image processing

Suitable for 2D dimensional and geometric inspection

Improves repeatability and reduces operator measurement errors

Commonly used in automotive, electronics, plastics, tooling, precision engineering, and quality-control laboratories



COORDINATE MEASURING MACHINE

A Coordinate Measuring Machine (CMM) is a high-precision inspection system used to measure the size, position, shape, and geometric accuracy of manufactured components in three dimensions.

The machine uses a precision probe to collect measurement points along the X, Y, and Z axes, while advanced software compares the measured data with engineering drawings or CAD models.

- Measures length, diameter, distance, angle, position, flatness, straightness, parallelism, and perpendicularity
- Performs accurate 3D dimensional and geometric inspection
- Supports contact and non-contact probing systems, depending on the machine configuration
- Provides high accuracy, repeatability, and traceable inspection results
- Suitable for first-piece inspection, production quality control, component validation, and calibration
- Widely used in automotive, aerospace, tool & die, precision engineering, and manufacturing industries

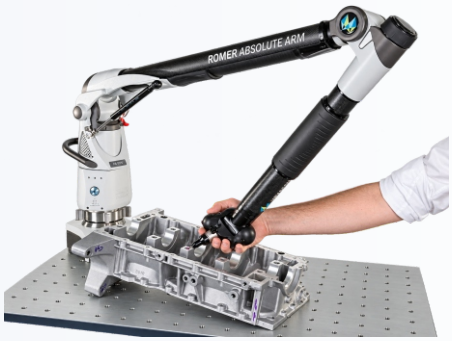
SALES & analyses

We provide complete after-sales support for 1D & 2D Height Gauges, VMMs, and CMMs, including installation, calibration, maintenance, troubleshooting, and technical assistance.
Our analyses ensures reliable performance, measurement accuracy, minimum downtime, and long-term equipment efficiency.

OUR REPORTING STANDARDS

1. I1SO Standards (International Organization for Standardization) ISO 230 Series — Machine tool test code (Geometric & positioning accuracy) ISO 230-2 - Positioning accuracy & repeatability ISO 230-4 - Circular test (Ballbar analysis)
2. VDI Standards (Verein Deutscher Ingenieure) VDI/DGQ 3441 - Statistical evaluation of machine tool accuracy VDI 2617 — Accuracy of coordinate measuring systems VDI 2852 — Machine tool performance testing
3. JIS Standards (Japanese Industrial Standards) JIS B 6201 — Machine tool testing methods JIS B 6330 — Machining center accuracy inspection JIS B 6190 - Ballbar testing methodology

OUR INSPECTIONS analyses

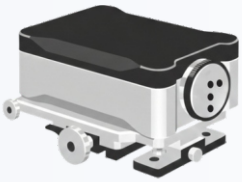


PORTABLE ARM INSPECTION

Portable Arm Inspection is an on-site dimensional measurement analyses using a portable Coordinate Measuring Arm to accurately inspect complex components, assemblies, and machined parts.

It helps verify dimensions, geometry, alignment, profiles, hole positions, GD&T features, and part-to-CAD comparison directly at the customer's location, reducing inspection time and avoiding the need to move heavy components.

IN ASSOCIATION WITH



Reddot
CALIBRATION SOLUTION
PRECISION • RELIABILITY • EXCELLENCE

GMS
GLOBAL METROLOGY
SERVICES


24×7 Support


On-Site Service


Rapid Response

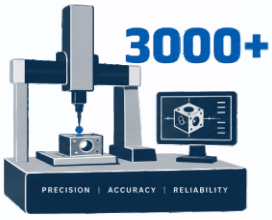

Preventive Maintenance


Technical Reporting


Post-Service Support



10 Years Experience Technicians



3000+ Machine Calibrated


Pan India Support

HEXACORE
PRECISION TECHNOLOGIES
www.hexacoreprecision.com

in associates with

Bengaluru, KA
No.1109 /1, 4th Floor,
HSR Layout, Sector 3,
22nd Cross Road,
Bengaluru - 560102.
M : 73390 15820
E : info@hexacoreprecision.com

Coimbatore, TN
57-A, 1st Floor, Smart Spaces,
KS Ramasamy Street,
Saibaba Colony,
Coimbatore - 641 037.
M : 86672 68799
E : reddotcalibration@gmail.com

Chennai, TN
Global Metrology analyses
No.4, Bharati Street,
Dhanalakshmi Nagar,
Avadi, Chennai - 600054.
M : 90944 79695
E : globalmetrologyanalysess@gmail.com